

Overview

`src/app/docsExploreFilters.ts` (added R1-158) introduces a filtering module for the documentation explorer. It defines symbol-kind groups, change-state filters, and utilities for toggling, matching, and mapping graph detail levels.

Key changes

- Import: `import type { CodeCityChangeState } from "./types";` (R1).
- Symbol kind group type and constants:
 - `export type SymbolKindGroup = "file" | "class" | "function";` (R4).
 - `ALL_SYMBOL_KIND_GROUPS` array (R6-10).
- Change-state filter constants: `ALL_CHANGE_STATE_FILTERS` (R12-17).
- Default helpers: `defaultSymbolKindGroups()` (R31-33) and `defaultChangeStateFilters()` (R35-37).
- Kind resolution: `symbolKindGroupForKind()` (R39-49).
- Match checks: `symbolMatchesKindGroups()` (R53-64) and `changeStateMatchesFilters()` (R68-76).
- Toggle utilities: `toggleSymbolKindGroup()` (R78-92) and `toggleChangeStateFilter()` (R94-108).
- Active-state predicates: `symbolKindsFilterActive()` (R110-115) and `changeStatesFilterActive()` (R117-124).
- Graph detail mapping: `graphDetailFromSymbolKinds()` (R127-151).
- Filter key generation: `exploreFilterKey()` (R153-158).

Impact

- **Correctness** – Functions provide deterministic filtering logic; no existing code paths are altered.
- **Maintainability** – Centralized constants and helpers reduce duplication across the explorer.
- **Performance** – Operations use small `Set`s and array filters; negligible runtime cost.
- **Compatibility** – Only new exports; existing modules remain unchanged.

Risks & follow-ups

1. **Toggle edge cases** – Verify that toggling with an empty array defaults to all groups/states and that single-item removal preserves the array.
2. **Graph detail logic** – Ensure `graphDetailFromSymbolKinds()` returns the expected `detailLevel` and `showFunctionsInClassView` for all combinations of enabled groups.
3. **Filter key ordering** – Confirm that `exploreFilterKey()` consistently sorts keys to avoid cache misses.

4. **Lint & build** – Run the repo's lint, test, and production build to catch any type or import errors introduced by the new file.